

ActiveScale: Scaling Active Perception for Robots across Model, Data, and Hardware

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Fig. 1: Overview of ActiveScale: (1) egocentric human data provide a scalable path for transferring active-perception priors to robots through human-robot co-training, (2) Active-perception Mobile-manipulation Platform (AMP) enables coordinated active-perception and mobile-manipulation teleoperation, while pose-grounded temporal modeling supports cross-view reasoning and action generation, and (3) real-world evaluation assesses these capabilities across five manipulation tasks and a mobile-manipulation demonstration.

Abstract—Active perception is essential for robotic manipulation when fixed viewpoints leave task-relevant information occluded or unobserved. However, enabling vision-language-action (VLA) models to reason across changing viewpoints and actively acquire informative observations remains challenging. We present ActiveScale, a framework that advances active perception through coordinated model, data, and hardware designs. Our model augments a VLA with historical video observations and explicit camera-pose supervision, using per-frame pose tokens and a lightweight prediction head to associate observations across viewpoints and support a coherent understanding of the scene. To learn from the camera motion naturally present in human activity, we introduce a scalable human-robot mid-training recipe using 1000 hours of egocentric and robotic data, adapting the model to temporal inputs and pose supervision. We further introduce Active-perception Mobile-manipulation Platform (AMP), a robotic platform that supports active perception and mobile manipulation through single-operator teleoperation, enabling scalable collection of demonstrations that coordinate viewpoint changes and manipulation. Experiments demonstrate improved success rates on active-perception tasks, while ablation studies validate the contributions of camera-pose-aware modeling and egocentric mid-training. Together, these components provide an integrated foundation for studying and developing active perception in

robotic manipulation. More information and materials are available at active-scale.github.io.

I. INTRODUCTION

Recent advances in robotic foundation models, such as vision-language-action models (VLAs) [1]–[4], have substantially broadened robots’ ability to perform diverse tasks across environments. However, as robotic settings become richer and tasks more complex, observations from a fixed viewpoint are often only partially informative. Successfully completing a task may require the robot to actively acquire task-relevant information in cluttered environments—for example, by reorienting its egocentric view to inspect a bowl hidden inside a cabinet or the contents of a backpack, thereby reducing occlusion. Consequently, static perception is increasingly insufficient, making **active perception** a critical capability for general-purpose robotic agents.

Prior work has explored several directions toward active perception. SAPAVE [5] decouples the training of camera motion and arm motion, enabling effective learning of both capabilities. ActiveMimic [6] leverages egocentric data for active-perception pretraining, while EgoMI [7] introduces

a hardware platform for scaling egocentric demonstration collection with approximately 200 hours of data for policy training. Despite this progress, three key challenges remain. First, accomplishing key active-perception tasks requires a model to infer its camera pose and predict subsequent camera motion, with the latter contingent on the former. However, existing VLAs lack these capabilities. Second, although egocentric data naturally contains camera motion, existing methods do not effectively exploit this signal for learning active perception. Third, current approaches predominantly study active perception in static environments [5], [6], leaving a substantial gap to real-world mobile manipulation, where both the robot and its observation viewpoint must adapt to dynamic, cluttered settings. *More importantly, a unified infrastructure for systematically studying active perception across model, data, and hardware remains lacking.*

To address the aforementioned challenges, we propose **ActiveScale**. It consists of 3 important designs: (1) **Model**. To enable perception of a coherent 3D world, we first extend the input from a single image to a video sequence containing historical frames. We then introduce a simple task: a learnable camera token follows each front-camera frame’s visual features, and a linear projector and auxiliary camera head predict translation, rotation, and field of view from the corresponding VLM hidden states. This design enables the VLA not only to explicitly understand the camera pose of the current frame, but also to associate video observations across different frames, thereby allowing it to perceive the persistent scene across viewpoints using only a sequence of consecutive video frames as input. (2) **Data and mid-training**. To adequately train the model to learn camera motion, we mid-train the model described in (1) on large-scale egocentric data and robot data, totaling over 1000 hours. This provides a scalable training recipe that adapts the pretrained model to the augmented inputs and supervision while improving its understanding of camera motion. (3) **Hardware**. We develop an Active-perception Mobile-manipulation Platform (**AMP**) that integrates bimanual manipulation, an independently actuated camera, and mobile-base control through a unified Meta Quest 2 teleoperation interface. A single operator can jointly control viewpoint, manipulation, and base motion, providing a scalable interface for collecting coordinated mobile-manipulation demonstrations.

Diverse and sufficient experiments demonstrate that our proposed model achieves higher success rates on active perception tasks. Moreover, ablation studies validate the effectiveness of our model design and mid-training procedure. In summary, our contributions are threefold:

- We propose a pose-grounded temporal VLA for active perception and cross-view reasoning, achieving state-of-the-art performance across five real-world active-perception tasks.
- We propose a human–robot mid-training recipe using large-scale egocentric human and robot data. To the best of our knowledge, it is the first approach to scale active-perception training to 1000 hours of data—more than 5× larger than prior efforts. This recipe provides a scalable

path for transferring human manipulation and active-view priors to robot policies.

- We develop AMP for active perception and mobile manipulation, enabling single-operator scalable collection of demonstrations that jointly coordinate viewpoint, bimanual manipulation, and mobility.

II. RELATED WORK

A. Active Perception in Robot Manipulation

Active perception has long been studied in robotics and computer vision [8], [9], with classical approaches often formulated as Next-Best-View (NBV) planning [10]–[12]. These methods typically optimize viewpoint selection separately from downstream manipulation. Recent learning-based methods instead couple viewpoint control with manipulation. ViA [13], AV-ALOHA [14], and Eye, Robot [15] learn task-driven viewpoint control together with manipulation, while EgoMI [7] and ActiveUMI [16] exploit coordinated human viewpoint and manipulation demonstrations. VLA-based approaches further model camera control within visuomotor policies: SaPaVe [5] learns semantic camera control with decoupled camera and manipulation actions, whereas ActiveMimic [6] models camera and manipulation motion jointly in a unified action space. Our work follows this direction while explicitly grounding temporal camera motion in the VLA representation.

B. Learning from Egocentric Human Videos

Egocentric human videos provide a scalable source of diverse embodied experience for robot learning beyond robot-collected data. EgoMimic [17], EgoBridge [18], and EgoScale [19] align human manipulation trajectories with robot embodiments for policy transfer, while Kareer et al. [20] show that human-to-robot transfer can emerge through joint training with sufficiently diverse robot pretraining. Importantly, egocentric videos encode more than hand and wrist motion: human head-camera trajectories reflect task-driven visual intent, as people actively reposition their viewpoints to search for relevant objects, resolve occlusions, and support subsequent manipulation. Recent works including EgoMI [7] and ActiveUMI [16] exploit this coordinated head–hand behavior to learn active perception together with manipulation. Our work builds on this perspective, using egocentric human data to transfer both manipulation priors and task-driven viewpoint behaviors to robots.

C. Spatial Intelligence for VLA

Existing methods enhance VLA spatial understanding through geometric inputs or auxiliary supervision. PointVLA [21], StereoVLA [22], and SpatialVLA [23] incorporate point clouds, stereo cues, and depth-derived spatial information, respectively. Spatial Forcing [24] first aligns visual representations with pretrained 3D features, while LingBot-VLA [25] and DualCoT-VLA [26] distill geometric knowledge into learnable queries. Inspired by reconstruction-based VLAs [27], GEM-VLA [28] and DepthVLA [29] reconstruct depth from RGB observations as an auxiliary

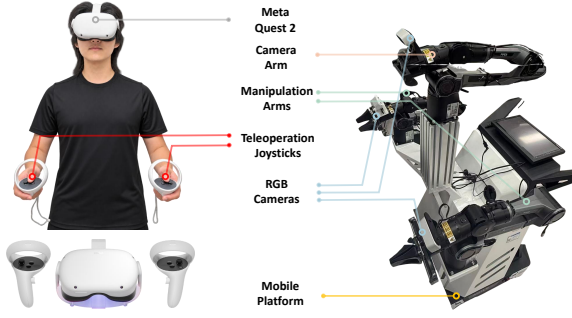


Fig. 2: AMP system overview. *Left*: Single-operator Meta Quest 2 interface. *Right*: AMP triple-arm robotic platform.

task. For active perception tasks, the central challenge is not to reconstruct 3D geometry, but to understand how camera motion within a persistent scene supports task execution. Accordingly, our ActiveScale incorporates historical frames to model camera-pose evolution and predicts future observations conditioned on image and pose histories.

III. METHODOLOGY

An overview of ActiveScale is shown in Fig. 1. In this section, we first formulate the active-perception manipulation problem (Sec. III-A), then introduce our robotic platform and teleoperation setup (Sec. III-B), followed by the model architecture (Sec. III-C) and training recipe (Sec. III-D).

A. Problem Formulation

We formulate active-perception manipulation as learning a language-conditioned policy π_θ that jointly controls manipulation and viewpoint motion. At time t , given an observation $\mathbf{o}_t \in \mathcal{O}$ and a language instruction $\ell \in \mathcal{L}$, the policy predicts a future action chunk $\mathbf{a}_{t:t+K-1} \in \mathcal{A}^K$, i.e., $\pi_\theta(\mathbf{o}_t, \ell) \mapsto \mathbf{a}_{t:t+K-1}$. The observation $\mathbf{o}_t$ comprises visual observations and a state context $\mathbf{s}_t$ containing non-visual robot information such as proprioception and kinematics. Each action $\mathbf{a}_\tau$ jointly specifies the Cartesian poses of the left and right manipulation end effectors and the active camera, together with the two gripper commands, all expressed in a shared robot-centric frame $\mathcal{F}_U$.

B. AMP Setup

Our active-perception mobile-manipulation platform (AMP) is built on a modified AgileX Cobot-Magic, as shown in Fig. 2. The original platform consists of a differential-drive mobile base and two 6-DoF Piper arms, to which we add a third arm for active viewpoint control. The left and right arms perform bimanual manipulation and are each equipped with an Orbbec DaBai DC1 camera near the end effector. To enable active viewpoint control, we replace the original fixed front-facing camera with an additional AgileX 6-DoF Piper arm carrying a third Orbbec DaBai DC1 camera at its end effector. This design provides an independently actuated egocentric viewpoint while preserving both manipulation arms and the mobile base for task execution.

1) *Teleoperation System*: To enable a single operator to jointly control the active viewpoint, bimanual manipulation, and mobile base, AMP uses a Meta Quest 2 headset and two handheld controllers as the teleoperation interface. At time t , we obtain the 6-DoF poses $\mathbf{T}_t^S = (\mathbf{R}_t^S, \mathbf{p}_t^S) \in SE(3)$, where $S \in \{H, L, R\}$ denotes the headset, left controller, and right controller, respectively. The headset controls the middle arm carrying the active camera C , while the left and right controllers control the corresponding manipulation arms. We transform each tracked pose from the Quest tracking frame to the robot control convention using a fixed axis transformation. Before teleoperation, the three arms are initialized to predefined configurations that approximately match the relative layout of the operator’s head and hands. When the operator engages teleoperation by pressing both controller grip buttons, we register the current tracked poses $\mathbf{T}_0^S$ together with the current robot end-effector poses $\mathbf{T}_0^A$ as reference configurations. Subsequent human motion is expressed relative to these references as $\Delta \mathbf{p}_t^S = \mathbf{p}_t^S - \mathbf{p}_0^S$ and $\Delta \mathbf{R}_t^S = \mathbf{R}_t^S (\mathbf{R}_0^S)^\top$. The corresponding robot target is then computed as $\mathbf{p}_t^A = \mathbf{p}_0^A + \lambda \Delta \mathbf{p}_t^S$ and $\mathbf{R}_t^A = \Delta \mathbf{R}_t^S \mathbf{R}_0^A$, with $\lambda = 1$, and tracked through inverse kinematics. Re-engaging teleoperation re-anchors these references, allowing the operator to reposition the headset and controllers without discontinuities. The controller triggers continuously control the left and right grippers. For mobile control, the left-controller X/Y buttons rotate the base left/right, while the right-controller A/B buttons drive it backward/forward, allowing the operator to navigate the workspace without physically following the robot. Thus, a single operator can coordinate camera motion, bimanual manipulation, gripper control, and platform navigation within the same interface.

2) *Data Collection*: We record synchronized demonstrations at 30Hz, including the three RGB streams $\mathbf{I}_t = (I_t^f, I_t^l, I_t^r)$, camera intrinsics (f_x, f_y, c_x, c_y) , distortion parameters, image resolution, and joint feedback and control commands for all three arms. To obtain a representation that remains local to the robot as the mobile base moves, we express the realized poses of both manipulation end effectors and the active camera in a shared robot-centric frame $\mathcal{F}_U$ rigidly attached to the mobile platform. Specifically, $\mathcal{F}_U$ shares the orientation of the middle-arm base frame, with its origin shifted 0.25m downward along the middle-arm z axis. Thus, $\mathcal{F}_U$ moves together with the robot rather than remaining fixed at the platform’s initial world pose. For $X \in \{L, R, C\}$, corresponding to the left end effector, right end effector, and active camera, respectively, we represent its pose as ${}^U \xi_t^X = [{}^U \mathbf{p}_t^X, {}^U \mathbf{q}_t^X] \in \mathbb{R}^7$, where ${}^U \mathbf{p}_t^X \in \mathbb{R}^3$ denotes position in $\mathcal{F}_U$ and ${}^U \mathbf{q}_t^X$ is an $xyzw$ -ordered unit quaternion. Together with the two gripper positions, this gives the 23-D robot-centric state $\mathbf{s}_t = [{}^U \xi_t^L, {}^U \xi_t^R, {}^U \xi_t^C, \gamma_t^L, \gamma_t^R]$. We use the corresponding realized Cartesian trajectories and teleoperated gripper commands as pose-space action targets in the same representation. We separately record the planar motion of the mobile base as $\mathbf{b}_t = (v_t, \omega_t)$, where v_t and ω_t denote the forward linear and yaw angular velocities, respectively.

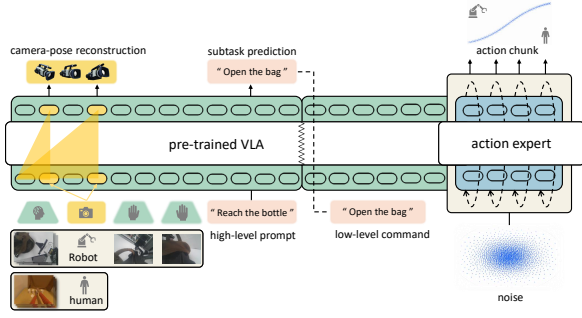


Fig. 3: Model overview. A pretrained VLA backbone encodes visual, language, and state context. Pose-grounded camera tokens support camera-pose reconstruction, while the VLM predicts semantic outputs such as subtasks and a flow-matching action expert generates manipulation and active-camera actions. The architecture is shared across human and robot training data.

C. Model Architecture and Objectives

Active perception requires a policy to relate visual observations across changing viewpoints while jointly reasoning about camera motion and manipulation. As shown in Fig. 3, we build on $\pi_{0.5}$ [1] and augment its visual context with pose-grounded camera tokens inspired by Cambrian-P [30], while treating camera motion as part of the action space.

1) $\pi_{0.5}$ Backbone: We adopt $\pi_{0.5}$ [1] as our base VLA architecture. $\pi_{0.5}$ consists of a pretrained PaliGemma [31] vision-language model (VLM) and a lightweight transformer-based action expert. The VLM encodes multi-view visual observations, language instructions, and tokenized proprioceptive states to provide context for action generation. Conditioned on this context, the action expert predicts continuous action chunks using flow matching. Given a clean action chunk $\mathbf{a}$, Gaussian noise $\epsilon \sim \mathcal{N}(0, I)$, and flow time $\tau \in [0, 1]$, we construct $\mathbf{x}_\tau = \tau\epsilon + (1 - \tau)\mathbf{a}$ with target velocity $\mathbf{u}_\tau = \epsilon - \mathbf{a}$. The action expert predicts a conditional velocity field $\mathbf{v}_\theta(\mathbf{x}_\tau, \tau, \mathbf{c})$, where $\mathbf{c}$ denotes the VLM context, and is trained with $\mathcal{L}_{\text{flow}} = \mathbb{E} \left[\|\mathbf{v}_\theta(\mathbf{x}_\tau, \tau, \mathbf{c}) - \mathbf{u}_\tau\|_2^2 \right]$. In addition to flow-matching supervision, we retain the autoregressive token-prediction objectives of $\pi_{0.5}$. At the high level, the VLM predicts the current subtask from the episode-level instruction and visual-state context, supervised with a standard next-token cross-entropy loss $\mathcal{L}_{\text{subtask}}$. At the low level, the current subtask serves as the action-conditioning instruction. FAST [32] converts the future continuous action trajectory into discrete action tokens, which are autoregressively predicted with next-token cross-entropy loss $\mathcal{L}_{\text{FAST}}$. The same subtask-conditioned context also conditions the continuous flow-matching action expert.

2) Camera Tokens: To provide spatial context across changing viewpoints, we augment the $\pi_{0.5}$ input with historical observations that retain previously observed scene information. We associate each frame with a learnable camera token [30], supervised to predict its camera pose and field of view, encouraging the model to relate observations across viewpoints. We use the current front-camera observation together with three past observations sampled at 16-frame intervals, denoted by $\mathcal{H}_t^f = (I_{t-48}^f, I_{t-32}^f, I_{t-16}^f, I_t^f)$. Each

frame is encoded by the shared vision encoder, and a learnable camera token is appended after its visual tokens. Specifically, we use two learnable embeddings $\mathbf{c}_{\text{first}}$ and $\mathbf{c}_{\text{rest}}$ for the earliest valid frame and subsequent frames, respectively. The visual-language prefix is therefore organized as $[\mathbf{V}_{t-48}^f, \mathbf{c}_1, \dots, \mathbf{V}_t^f, \mathbf{c}_4, \mathbf{V}_t^l, \mathbf{V}_t^r, \mathbf{E}(\ell, \mathbf{s}_t)]$, where $\mathbf{V}$ denotes visual tokens and $\mathbf{E}(\ell, \mathbf{s}_t)$ denotes the tokenized language and proprioceptive-state context. We apply block-causal attention over the front-camera history: each temporal block can attend to itself and preceding history, while its camera token cannot access future frames. The subsequent wrist, language, and action tokens can attend to all valid camera-history blocks.

The hidden state of each camera token is linearly projected to a VGGT-style camera head [33], which predicts a 9-D camera representation consisting of 3D translation (x, y, z) , quaternion orientation (q_x, q_y, q_z, q_w) , and vertical and horizontal fields of view $(\text{fov}_y, \text{fov}_x)$. For camera tokens with valid geometric targets, we supervise the camera head with

$$\mathcal{L}_{\text{cam}} = \mathcal{L}_{\text{trans}} + \mathcal{L}_{\text{rot}} + \lambda_{\text{fov}} \mathcal{L}_{\text{fov}}, \quad (1)$$

where the three terms are component-wise ℓ_1 losses on translation, quaternion orientation, and field of view, respectively, and we use $\lambda_{\text{fov}} = 0.5$. This auxiliary head geometrically grounds the camera-token representation and is removed during action inference. The resulting prefix conditions the original $\pi_{0.5}$ action expert, which jointly generates manipulation and head camera motions.

D. Training Recipe

Starting from the pretrained $\pi_{0.5}$ checkpoint [1], which provides broad visual-semantic and robot-control priors from large-scale pretraining, we train our policy in two stages. We first perform human-robot mid-training, leveraging diverse egocentric human demonstrations to acquire manipulation and active-view priors while interleaving robot demonstrations to maintain robot-aligned action learning. We then perform task-specific post-training on our teleoperation data to adapt the policy to the target embodiment and tasks.

1) Human-Robot Co-training: Egocentric human data provide scalable supervision for transferable manipulation priors [19]. Recent work further shows that human-to-robot transfer can emerge through human-robot co-training with a sufficiently diverse pretrained VLA [20]. Meanwhile, continuous action adaptation can interfere with semantic knowledge inherited from the pretrained VLM [34]. Motivated by these observations, we jointly train on human and robot demonstrations, leveraging human data to acquire transferable semantic, manipulation, and active-view priors while retaining robot-aligned action learning. Human and robot demonstrations are sampled at a balanced 1:1 ratio. For human demonstrations, head-camera poses provide active-view supervision and wrist trajectories provide manipulation supervision. Following [6], we re-express the human camera and wrist trajectories in the camera frame of the earliest valid observation in each temporal window, while robot trajectories remain in their fixed robot reference frames. We

normalize states and action trajectories separately for each data source and mask unavailable action dimensions and camera-pose targets.

The human data are drawn from the large-scale and diverse egocentric datasets EgoLive [35] and EgoVerse [36], which provide rich first-person manipulation experience together with head-camera motion. We further curate these datasets by retaining indoor household-manipulation clips with valid camera trajectories and the task/action supervision required by our training pipeline, while prioritizing activities semantically related to our downstream settings. The robot portion combines diverse demonstrations from AgiBot World [37], [38], RoboCOIN [39], and our AMP, covering diverse robot embodiments and control distributions.

The mixed-batch training objective is $\mathcal{L}_{co} = \mathcal{L}_{subtask}^H + \lambda_H(\mathcal{L}_{FAST}^H + \mathcal{L}_{flow}^H) + \mathcal{L}_{FAST}^R + \mathcal{L}_{flow}^R + \lambda_C \mathcal{L}_{cam}$. Here, H and R denote human and robot examples, and each term is evaluated only when the corresponding supervision is available in the data. High-level human samples supervise $\mathcal{L}_{subtask}^H$, while low-level human and robot samples supervise their FAST and flow objectives. We use $\lambda_H = \lambda_C = 0.2$.

2) *Post-training with Teleoperation Data*: After mid-training, we perform task-specific post-training on the AMP teleoperation demonstrations collected with the system described above. We keep the model architecture and action representation unchanged and optimize $\mathcal{L}_{post} = \mathcal{L}_{flow} + \lambda_C \mathcal{L}_{cam}$, with $\lambda_C = 0.2$. Unlike in human-robot co-training, we do not apply the autoregressive subtask or FAST objectives, since the post-training data directly provide aligned robot action and active-camera trajectories in the target embodiment. This stage specializes the co-trained policy to the AMP and downstream manipulation tasks.

IV. EXPERIMENTS

We train our models on NVIDIA H100 GPUs and deploy them on AMP for real-world evaluation, focusing on tasks in which successful execution requires the robot to actively acquire task-relevant visual information during manipulation. Our evaluation addresses four complementary questions:

- **Q1: Active-perception performance.** How well does our policy perform on real-world manipulation tasks that require active viewpoint control?
- **Q2: Human-robot transfer.** Does human-robot mid-training improve downstream robot performance?
- **Q3: Model design.** How do historical observations and pose-supervised camera tokens contribute to active-perception performance?
- **Q4: Execution efficiency.** Can our policy meet the requirements of real-time deployment?

We first introduce the real-world task suite and evaluation protocol. For a reproducible comparison, we evaluate against $\pi_{0.5}$, a state-of-the-art VLA for real-world robotic manipulation, as recent active-perception VLAs do not publicly release checkpoints and complete training recipes for direct evaluation. We then study the effects of human-robot mid-training and our camera-aware VLA model design through controlled ablations, and analyze corresponding execution

efficiency. Finally, we demonstrate AMP on a single-operator mobile-manipulation task that combines active viewpoint control, bimanual manipulation, and base motion.

A. Task Suite and Evaluation Protocol

Active perception is particularly important when task-relevant information is initially occluded or outside the robot’s field of view, or when the robot must search a larger workspace to localize relevant objects or targets. To cover these two settings, we design five real-world task families, illustrated in Fig. 4, that induce complementary forms of viewpoint demand.

- **Bag.** The robot retrieves a bottle from inside a backpack and places it on the table. The bottle is initially occluded, requiring the robot to inspect the backpack interior while opening it; success of this task requires the bottle to be stably placed on the table.
- **Drawer.** The robot opens the upper or middle drawer, retrieves the instructed object, places it on the plate, and closes the drawer. The target object becomes visible only after opening; success of this task requires the object on the plate and the drawer closed.
- **Table.** The robot retrieves an instructed object underneath the table and places it into the specified labeled bin. The target is initially outside the frontal view, requiring downward viewpoint adjustment; success of this task requires the object in the instructed bin.
- **Pot.** The robot retrieves the instructed vegetable from the shelf, places it in the pot, and covers the pot with the lid. The robot must shift its viewpoint across the shelf, pot, and lid; success requires the correct vegetable in the pot with the lid stably placed.
- **Box.** The robot retrieves an instructed object from the shelf and places it into the left or right target box according to the language instruction. Different shelf heights require vertical viewpoint search; success requires the object in the instructed target box.

Together, these tasks span container and articulated occlusion, below-table perception, multi-stage viewpoint transitions, and large vertical workspace variation. For each task family, we use 150 demonstrations for post-training and evaluate each method over 20 real-world rollouts under matched conditions. A rollout succeeds only if all required stages and the task-specific final-state criterion are satisfied. We report strict task success rate (SR) and task progress (TP). TP sums the credits for completed stages in Table I and is averaged over all rollouts for each task.

B. Main Experiments (Q1)

We compare the complete two-stage ActiveScale pipeline against the $\pi_{0.5}$ baseline under the same downstream post-training and evaluation protocol. The baseline is directly post-trained from the pretrained $\pi_{0.5}$ checkpoint on the task-specific demonstrations, whereas ActiveScale undergoes human-robot mid-training before post-training on the same data. As shown in Fig. 5, ActiveScale achieves higher SR and TP across all five task families, increasing mean SR

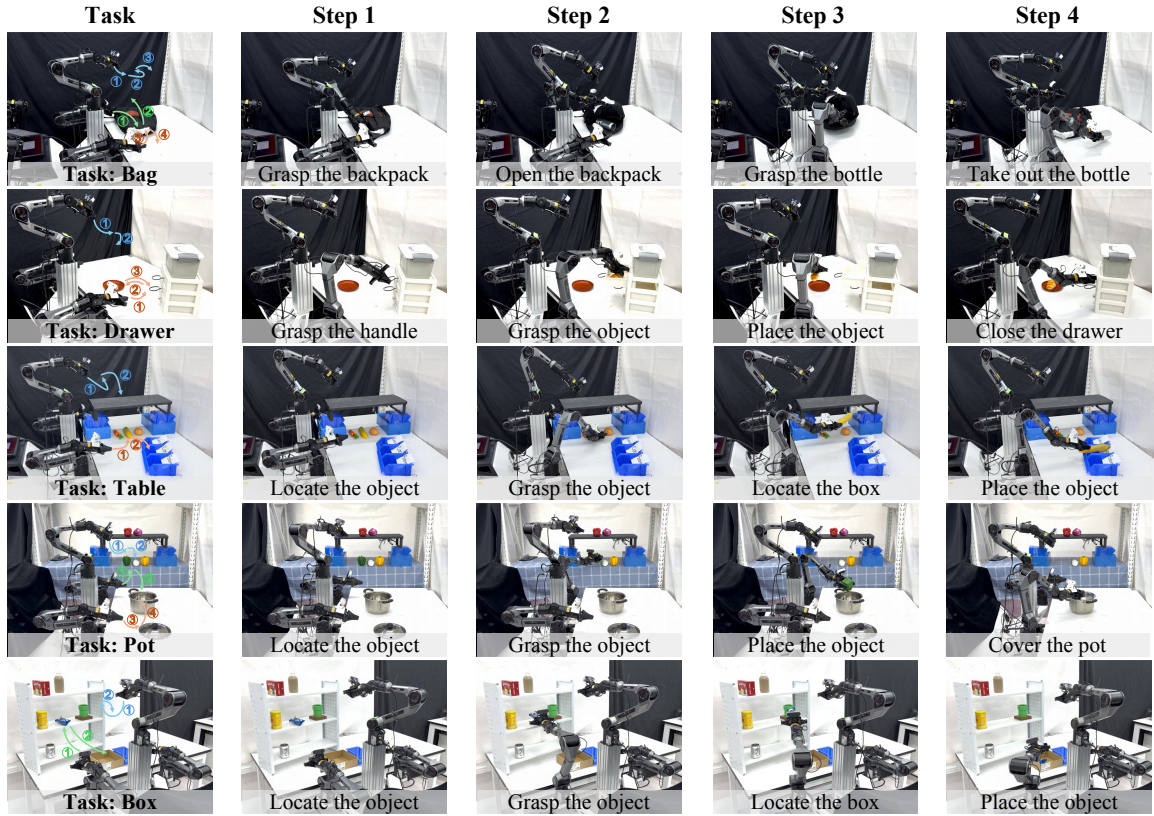


Fig. 4: Overview of the five real-world active-perception task families, grouped into two complementary settings: *Occlusion-Driven Manipulation* (Bag, Drawer, and Table), where task-relevant objects are initially hidden or outside the default view, and *Active Search and Localization* (Pot and Box), where the robot must actively search across workspace regions and localize instructed objects or targets.

TABLE I: Task progress scoring criteria. Parentheses indicate stage credit; dashes denote inapplicable stages.

Task	Stage 1	Stage 2	Stage 3	Stage 4
Bag	Grasp backpack (0.3)	Grasp bottle (0.3)	Place bottle on table (0.4)	—
Drawer	Open drawer (0.3)	Grasp target object (0.3)	Place object on plate (0.2)	Close drawer (0.2)
Table	Grasp target object (0.5)	Place object in target bin (0.5)	—	—
Pot	Grasp target vegetable (0.4)	Place vegetable in pot (0.2)	Grasp lid (0.2)	Cover pot (0.2)
Box	Grasp target object (0.5)	Place object in target box (0.5)	—	—

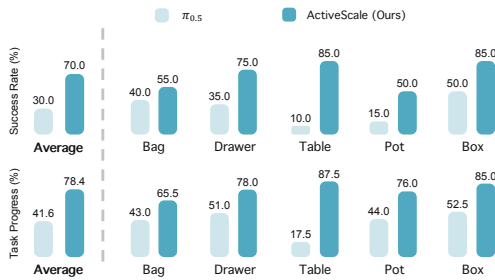


Fig. 5: Comparison with $\pi_{0.5}$ across five manipulation tasks.

from 30.0% to 70.0% and the mean TP from 41.6% to 78.4%. These gains indicate that the proposed model is substantially more effective at completing manipulation tasks in which useful visual information must be actively acquired during execution. In particular, the improvements are most pronounced when the target is initially hidden or outside the default view. On *Table*, $\pi_{0.5}$ can stall without initiating a grasp, whereas ActiveScale redirects its viewpoint, localizes the target, and proceeds with manipulation. Similarly, on *Bag*, ActiveScale actively inspects the backpack interior

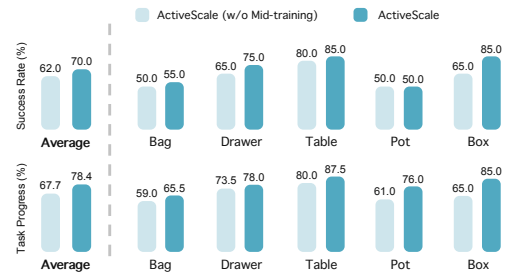


Fig. 6: Ablation study on the effect of mid-training.

while opening it, enabling subsequent object retrieval. Overall, these results show that ActiveScale more effectively coordinates viewpoint adjustment with manipulation and exploits newly acquired observations to advance task execution.

C. Ablation Studies

We next isolate the contributions of our training recipe and model design, corresponding to Q2 and Q3. Unless otherwise specified, all variants use the same task-specific post-training data and real-world evaluation protocol.

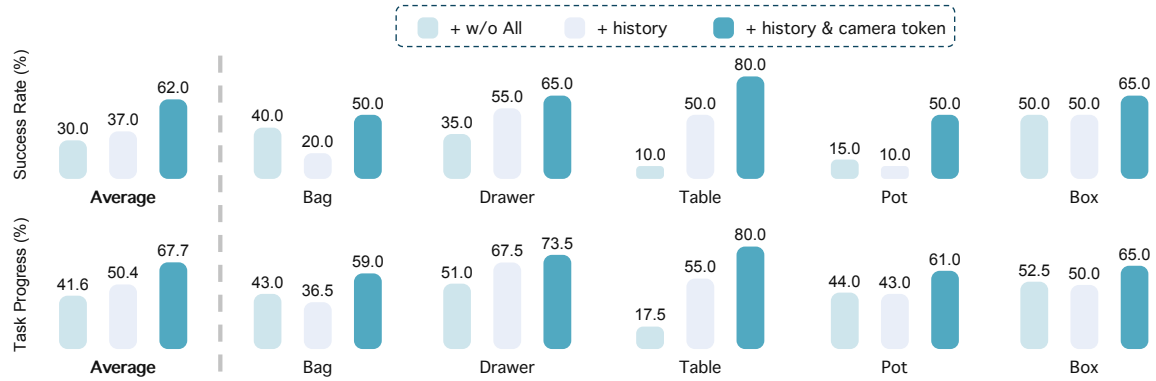


Fig. 7: Ablation study on history and camera token settings.

1) *Human–Robot Mid-Training (Q2)*: We isolate the effect of human–robot mid-training by comparing two variants of our model with the same architecture, task-specific demonstrations, and post-training protocol. The variant without mid-training is directly post-trained from the pretrained $\pi_{0.5}$ checkpoint, whereas the full model first undergoes human–robot mid-training and is then post-trained on the same downstream data. As shown in Fig. 6, human–robot mid-training improves overall downstream performance, although the magnitude of the gain varies across tasks. On *Pot*, for example, TP improves while SR remains unchanged. Qualitatively, the mid-trained policy produces better-aligned lid placements, whereas the variant without mid-training can misalign the lid and hesitate before release. This suggests that mid-training can improve intermediate execution quality even when the improvement is not fully reflected by binary task success. Overall, these results support our two-stage training recipe of human–robot mid-training followed by task-specific post-training.

2) *Temporal History and Camera Tokens (Q3)*: To isolate the contributions of temporal history and pose-grounded camera tokens from the effect of mid-training, we compare three variants using task-specific post-training only: original $\pi_{0.5}$, a history-only variant, and our full temporal representation with historical observations and pose-grounded camera tokens. As shown in Fig. 7, adding historical observations alone has a task-dependent effect, improving performance on *Drawer* and *Table* but not consistently across all task families. On *Bag*, for example, the history-only policy sometimes fails to open the backpack and performs worse than the single-frame base policy. These mixed results suggest that retaining past observations can provide useful temporal context, but history alone is insufficient to reliably interpret observations acquired from changing viewpoints. In contrast, adding pose-grounded camera tokens improves both SR and TP across all five task families relative to the history-only variant and recovers the performance drops observed on *Bag* and *Pot*. This consistent improvement supports explicitly grounding temporal observations with camera motion, helping the policy relate visual information acquired from different viewpoints.

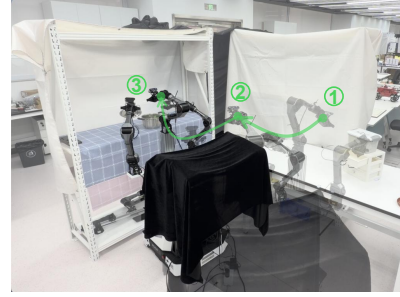


Fig. 8: Single-operator AMP teleoperation demonstration combining active perception and mobile manipulation. See the supplementary video for the full sequence.

D. Execution Efficiency (Q4)

To quantify our model’s inference efficiency, we adopt the action generation throughput metric used in OpenVLA-OFT [40]. With RTC [41] enabled, ActiveScale generates 50-action chunks with a throughput of 264.5 Hz on a single RTX 4090. Asynchronous execution overlaps inference with ongoing actions to eliminate waiting delays between action chunks, supporting continuous execution at the required control frequency of 30 Hz.

E. AMP Teleoperation Demonstration

To demonstrate the capability of AMP as a data-collection platform, we teleoperate a multi-stage mobile-manipulation task with a single operator, as shown in Fig. 8. The robot first actively adjusts its viewpoint to inspect a drawer and retrieve a vegetable, then redirects the camera to locate the cooking area before navigating the mobile base toward it. After reaching the destination, the operator completes the manipulation sequence by placing the vegetable into the pot and covering it with the lid. This demonstration shows that AMP enables a single operator to seamlessly coordinate active viewpoint control, bimanual manipulation, and mobile-base motion within one teleoperation interface.

V. CONCLUSIONS

We presented ActiveScale, a framework for active-perception robot learning across model, data, and hardware. ActiveScale augments a VLA with temporal observations and pose-grounded camera tokens, introduces a human–robot mid-training recipe using egocentric human and robot data,

and develops AMP, a single-operator platform for active viewpoint control, bimanual manipulation, and mobile-base motion. Across five real-world active-perception task families, ActiveScale outperforms directly post-trained $\pi_{0.5}$, improving mean task success from 30.0% to 70.0%, while ablations validate the benefits of human-robot mid-training and pose-grounded temporal modeling. Future work will scale AMP data collection and systematically evaluate learned policies on mobile active-perception tasks.

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